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TESTING  
CNAS L6451



## Guangdong Dongdian Testing Service Co., Ltd.

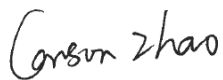
### TEST REPORT

|               |   |                                                                                                                                                                                                                                                                                                                                                                            |
|---------------|---|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Applicant     | : | Beijing InHand Networks Technology Co., Ltd.                                                                                                                                                                                                                                                                                                                               |
| Address       | : | Room 501, floor 5, building 3, yard 18, ziyue road, chaoyang district, Beijing                                                                                                                                                                                                                                                                                             |
| Manufacturer  | : | Beijing InHand Networks Technology Co., Ltd.                                                                                                                                                                                                                                                                                                                               |
| Address       | : | Room 501, floor 5, building 3, yard 18, ziyue road, chaoyang district, Beijing                                                                                                                                                                                                                                                                                             |
| Test sample   | : | InVehicle Router                                                                                                                                                                                                                                                                                                                                                           |
| Model No.     | : | VR624                                                                                                                                                                                                                                                                                                                                                                      |
| Test standard | : | EN 50155:2021                                                                                                                                                                                                                                                                                                                                                              |
| Test items    | : | Low temperature Test;<br>Dry heat Test;<br>Low temperature storage Test;<br>Cyclic damp heat Test;<br>Visual inspection;<br>DC Power supply test: supply voltage variations;<br>DC Power supply test: temporary supply overvoltages;<br>DC Power supply test: temporary supply undervoltage;<br>DC Power supply test: interruptions of supply voltage;<br>Insulation test; |
| Report No.    | : | DDT-RE26041734-1R01                                                                                                                                                                                                                                                                                                                                                        |
| Issue date    | : | 2026/06/26                                                                                                                                                                                                                                                                                                                                                                 |
| Issued By     | : | Guangdong Dongdian Testing Service Co., Ltd.<br>Unit 2, Building 1, No.17, Zongbu 2nd Road, Songshan Lake Park, Dongguan, Guangdong, China                                                                                                                                                                                                                                 |

# REPORT

## Test Report

|                      |                                                                                |                         |                       |
|----------------------|--------------------------------------------------------------------------------|-------------------------|-----------------------|
| Applicant:           | Beijing InHand Networks Technology Co., Ltd.                                   |                         |                       |
| Address:             | Room 501, floor 5, building 3, yard 18, ziyue road, chaoyang district, Beijing |                         |                       |
| Manufacturer:        | Beijing InHand Networks Technology Co., Ltd.                                   |                         |                       |
| Address:             | Room 501, floor 5, building 3, yard 18, ziyue road, chaoyang district, Beijing |                         |                       |
| Name of Test Sample: | InVehicle Router                                                               | Model No.:              | VR624                 |
| Identify Category:   | Commission test                                                                | Sample Delivery Method: | Enterprise sample     |
| Sample Receipt Date: | 2026/05/19                                                                     | Test Date:              | 2026/05/27~2026/06/04 |
| Test Standard:       | EN 50155:2021                                                                  |                         |                       |
| Test Items:          | See "Summary of test results"                                                  |                         |                       |
| Test Conclusion:     | See "Summary of Test Results" for details                                      |                         |                       |

|                                                                                     |                                                                                     |                                                                                       |
|-------------------------------------------------------------------------------------|-------------------------------------------------------------------------------------|---------------------------------------------------------------------------------------|
| Created: Yin Xusheng                                                                | Reviewed: Wang Bo                                                                   | Approved: Tony Yip                                                                    |
|  |  |  |
| 2026/06/20                                                                          | 2026/06/26                                                                          | 2026/06/26                                                                            |
| Created: Carson Zhao                                                                | Reviewed: Carl Huang                                                                | Approved: Henry Fu                                                                    |
|  |  |  |
| 2026/06/20                                                                          | 2026/06/25                                                                          | 2026/06/25                                                                            |

### Statement

1. This report is invalid without the special seal of the testing laboratory, partial copy report is invalid, and altered report is invalid.
2. The test results in this report are only responsible for the incoming samples, and the test results are only used by the client to understand the quality of the samples.
3. Without written consent of the laboratory, this report shall not be used as a commercial advertisement or as legal evidence.
4. If you have any objection to the test report, please submit it to the test laboratory within 15 days after receiving the report.
5. N/A: Not applicable.
6. The following tests are not included in the scope of CNAS accreditation: Visual inspection, DC Power supply test: supply voltage variations, DC Power supply test: temporary supply overvoltages, DC Power supply test: temporary supply undervoltage, DC Power supply test: interruptions of supply voltage,

Testing Laboratory: Guangdong Dongdian Testing Service Co., Ltd.

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Website: <http://www.dgddt.com/>

Revision History

| Version | Revision Content | Issue Date | Approved |
|---------|------------------|------------|----------|
| V0      | Initial issue    | 2026/06/26 | Tony Yip |
|         |                  |            |          |



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## 1. Summary of Test Results

| S/N: | Test Items                                            | Reference Standard | Sample State When Testing: | Test time                     | Sample #. | Test Results |
|------|-------------------------------------------------------|--------------------|----------------------------|-------------------------------|-----------|--------------|
| 1    | Low temperature Test                                  | EN 50155: 2021     | Power on                   | 2026/05/27<br>~<br>2026/05/28 | 3#        | PASS         |
| 2    | Dry heat Test                                         | EN 50155: 2021     | Power on                   | 2026/05/28<br>~<br>2026/05/30 | 3#        | PASS         |
| 3    | Low temperature storage Test                          | EN 50155: 2021     | Non-Operating              | 2026/05/30<br>~<br>2026/05/31 | 3#        | PASS         |
| 4    | Cyclic damp heat Test                                 | EN 50155: 2021     | Non-Operating              | 2026/06/02<br>~<br>2026/06/04 | 3#        | PASS         |
| 5    | Visual inspection                                     | EN 50155: 2021     | Non-Operating              | 2026/05/26<br>~<br>2026/06/02 | 2#        | PASS         |
| 6    | DC Power supply test: supply voltage variations       | EN 50155: 2021     | Operating                  | 2026/05/26<br>~<br>2026/06/02 | 2#        | PASS         |
| 7    | DC Power supply test: temporary supply overvoltages   | EN 50155: 2021     | Operating                  | 2026/05/26<br>~<br>2026/06/02 | 2#        | PASS         |
| 8    | DC Power supply test: temporary supply undervoltage   | EN 50155: 2021     | Operating                  | 2026/05/26<br>~<br>2026/06/02 | 2#        | PASS         |
| 9    | DC Power supply test: interruptions of supply voltage | EN 50155: 2021     | Operating                  | 2026/05/26<br>~<br>2026/06/02 | 2#        | PASS         |
| 10   | Insulation test                                       | EN 50155: 2021     | Non-Operating              | 2026/05/26<br>~<br>2026/06/02 | 2#        | PASS         |

## 2. Low temperature Test

### 2.1. Test Equipment

| S/N: | Equipment Name                 | Model       | Internal Number | Calibration Validity Period |
|------|--------------------------------|-------------|-----------------|-----------------------------|
| 1    | TEMP&HUMI Programmable chamber | ZXGDJS-225L | DDT-ZC00712     | 2026/09/09                  |
| 2    | DC Power supply                | SS-3020P    | DDT-ZC02830     | 2027/04/22                  |
| 3    | Programmable DC Power Supply   | IT6722A     | DDT-ZC02965     | 2026/10/22                  |

### 2.2. Test Conditions

#### Test 1

Low temperature: -30 °C

Temperature change rate:  $\leq 1$  °C/min

After the temperature stabilizes at -30 °C for no less than 2 hours, apply 48 V power supply and conduct functional inspection for no less than 0.5 hours.

Recovery time: no less than 1 hour.

#### Test 2

Low temperature: -30 °C

Temperature change rate:  $\leq 1$  °C/min

After the temperature stabilizes at -30 °C for no less than 2 hours, apply 9 V power supply and conduct functional inspection for no less than 0.5 hours.

Recovery time: no less than 1 hour.

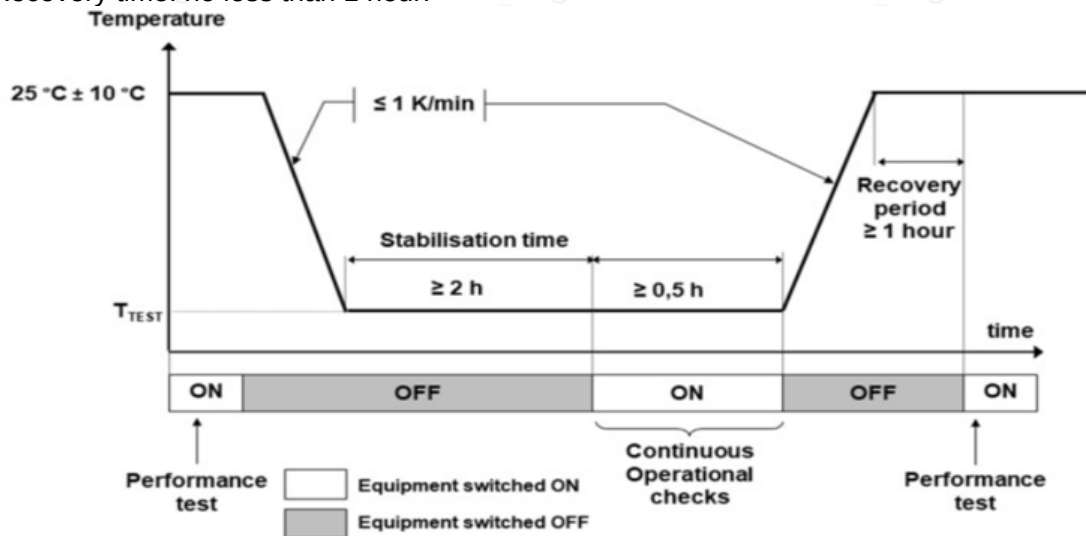


Figure 13 — Low temperature test

### 2.3. Test Requirement

Communication shall operate normally during and after the test.

### 2.4. Test Sample information

| Sample #. | Sample No.    | Sample Name      | Model No. |
|-----------|---------------|------------------|-----------|
| 3#        | S26041734-003 | InVehicle Router | VR624     |

## 2.5. Test Result

|                                                                 |                                  |
|-----------------------------------------------------------------|----------------------------------|
| Test site: Reliability Laboratory of Songshan Lake Headquarters | Site Ambient: 25.6°C; 53.3%RH    |
| Test Engineer: Yin Xusheng                                      | Test date: 2026/05/27            |
| Test standard: EN 50155: 2021                                   | Sample #: 3#                     |
| Operating modes: Power on                                       | Sample state during testing: 48V |
| Remark: /                                                       |                                  |

Operating voltage: 48 V. No functional abnormalities during and after the test.

|                                                                 |                                 |
|-----------------------------------------------------------------|---------------------------------|
| Test site: Reliability Laboratory of Songshan Lake Headquarters | Site Ambient: 25.1°C; 56.1%RH   |
| Test Engineer: Yin Xusheng                                      | Test date: 2026/05/28           |
| Test standard: EN 50155: 2021                                   | Sample #: 3#                    |
| Operating modes: Power on                                       | Sample state during testing: 9V |
| Remark: /                                                       |                                 |

Operating voltage: 9 V. No functional abnormalities during and after the test.

**Final test conclusion: PASS**



## 2.6. Test Photos

### 2.6.1. Test Photos



Sample appearance check (1)



Sample appearance check (2)



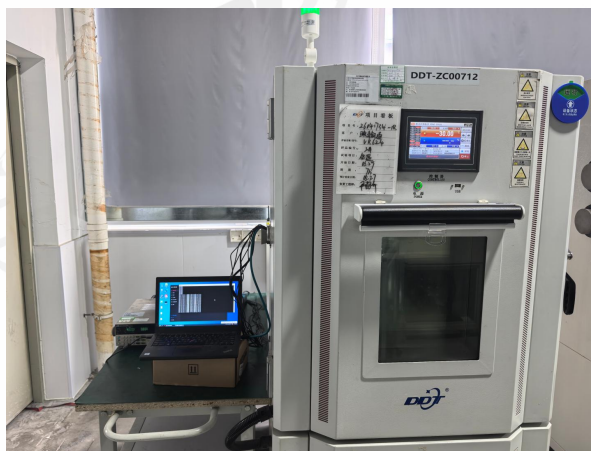
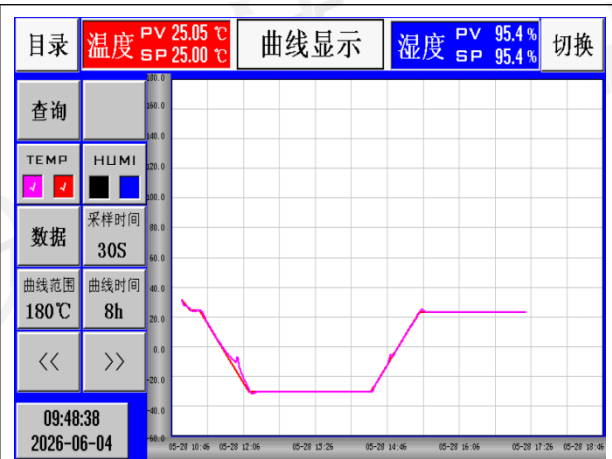
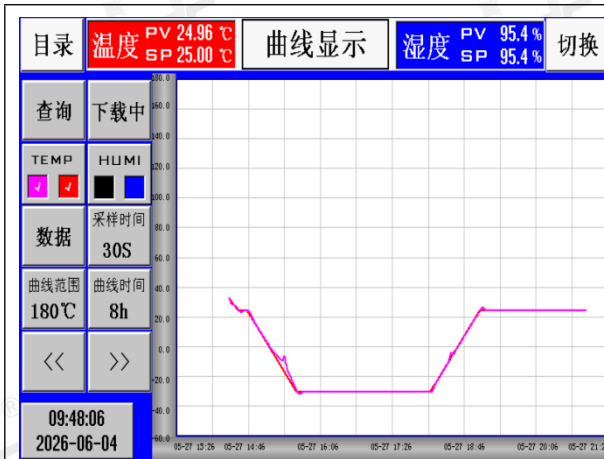
Functional Inspection of Samples Before Test



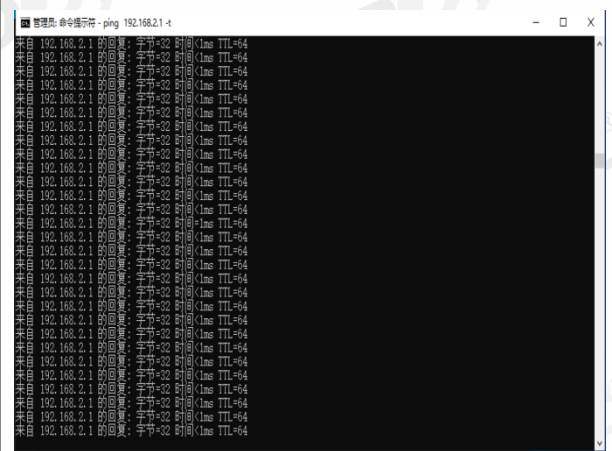
Sample placement



Test setup



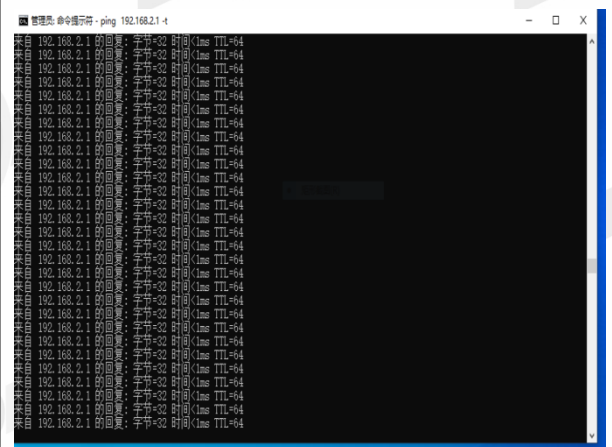
Functional Inspection During 48V Test (1)



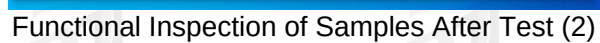
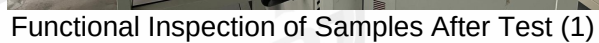
Functional Inspection During 48V Test (2)



Functional Inspection During 9V Test (1)



Functional Inspection During 9V Test (2)





### 3. Dry heat Test

#### 3.1. Test Equipment

| S/N: | Equipment Name                 | Model       | Internal Number | Calibration Validity Period |
|------|--------------------------------|-------------|-----------------|-----------------------------|
| 1    | TEMP&HUMI Programmable chamber | ZXGDJS-225L | DDT-ZC00929     | 2026/08/31                  |
| 2    | DC Power supply                | SS-3020P    | DDT-ZC02830     | 2027/04/22                  |
| 3    | Programmable DC Power Supply   | IT6722A     | DDT-ZC02965     | 2026/10/22                  |

#### 3.2. Test Conditions

Test 1

High temperature: 70 °C

Temperature change rate:  $\leq 1\text{ }^{\circ}\text{C/min}$

After the temperature stabilizes at 70 °C for no less than 2 hours, apply 55.2 V power supply and conduct functional inspection for no less than 6 hours. Then raise the temperature to 85 °C and perform functional inspection for no less than 10 minutes. Recovery time: no less than 1 hour.

Test 2

High temperature: 70 °C

Temperature change rate:  $\leq 1\text{ }^{\circ}\text{C/min}$

After the temperature stabilizes at 70 °C for no less than 2 hours, apply 10.35 V power supply and conduct functional inspection for no less than 6 hours. Then raise the temperature to 85 °C and perform functional inspection for no less than 10 minutes. Recovery time: no less than 1 hour.

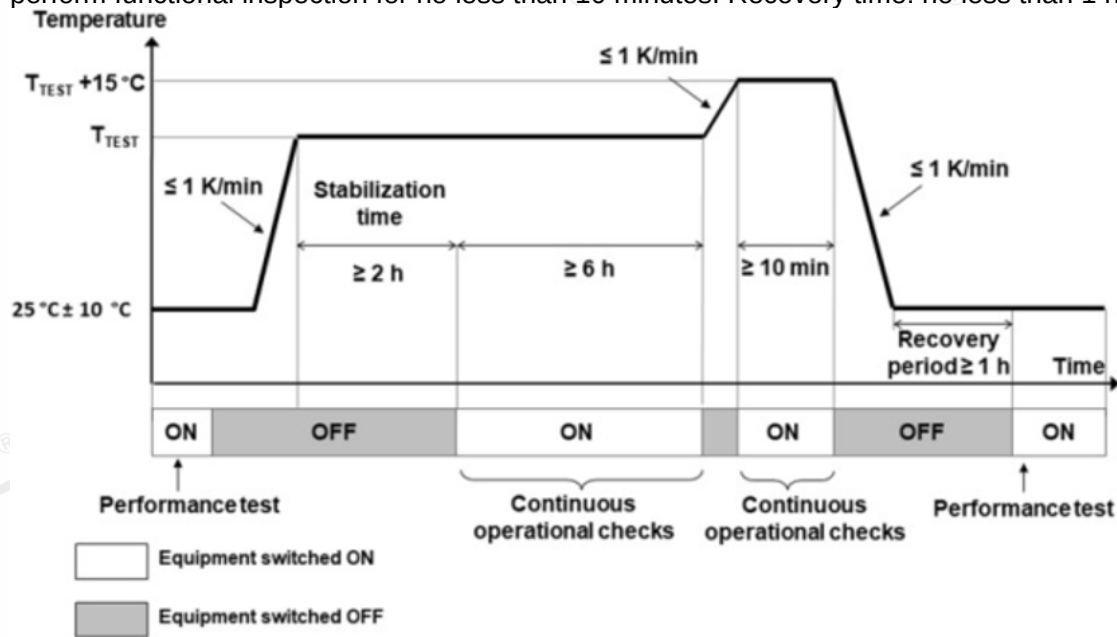


Figure 15 — Dry heat thermal test — Cycle B

#### 3.3. Test Requirement

Communication shall operate normally during and after the test.

#### 3.4. Test Sample information

| Sample #. | Sample No.    | Sample Name      | Model No. |
|-----------|---------------|------------------|-----------|
| 3#        | S26041734-003 | InVehicle Router | VR624     |

### 3.5. Test Result

|                                                                 |                                     |
|-----------------------------------------------------------------|-------------------------------------|
| Test site: Reliability Laboratory of Songshan Lake Headquarters | Site Ambient: 26.1°C; 51.2%RH       |
| Test Engineer: Yin Xusheng                                      | Test date: 2026/05/28~2026/05/29    |
| Test standard: EN 50155: 2021                                   | Sample #: 3#                        |
| Operating modes: Power on                                       | Sample state during testing: 10.35V |
| Remark: /                                                       |                                     |

Operating voltage: 10.35 V. No functional abnormalities during and after the test.

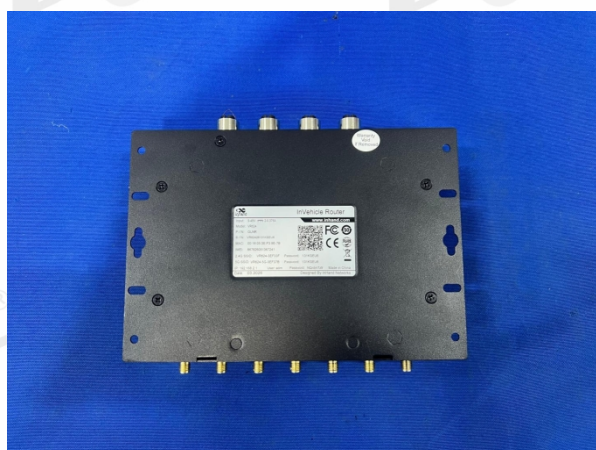
|                                                                 |                                    |
|-----------------------------------------------------------------|------------------------------------|
| Test site: Reliability Laboratory of Songshan Lake Headquarters | Site Ambient: 25.7°C; 52.9%RH      |
| Test Engineer: Yin Xusheng                                      | Test date: 2026/05/30              |
| Test standard: EN 50155: 2021                                   | Sample #: 3#                       |
| Operating modes: Power on                                       | Sample state during testing: 55.2V |
| Remark: /                                                       |                                    |

Operating voltage: 55.2 V. No functional abnormalities during and after the test.

**Final test conclusion: PASS**

### 3.6. Test Photos

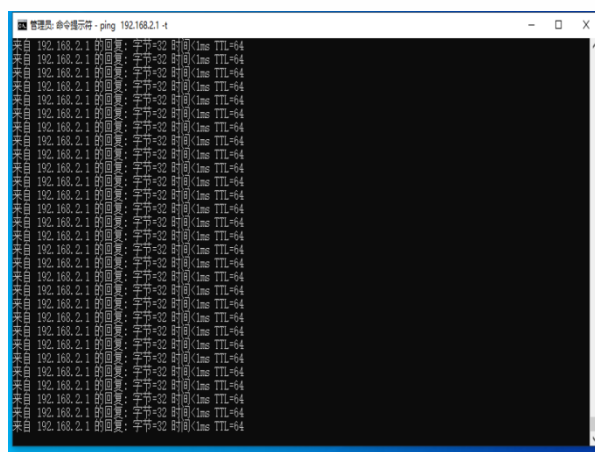
#### 3.6.1. Test Photos



Sample appearance check (1)



Sample appearance check (2)

Functional Inspection of Samples Before Test  
(1)Functional Inspection of Samples Before Test  
(2)

Sample placement

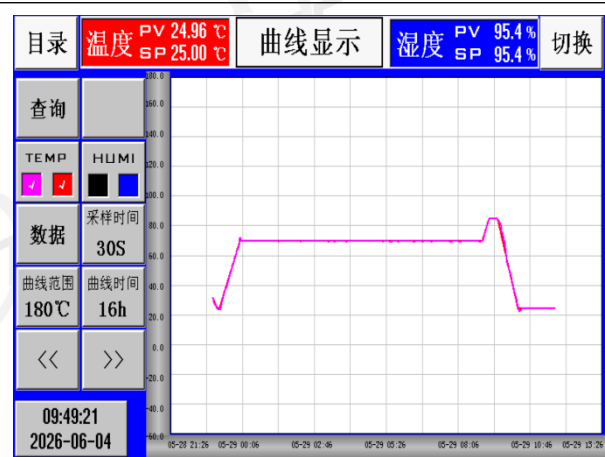


Test setup (1)

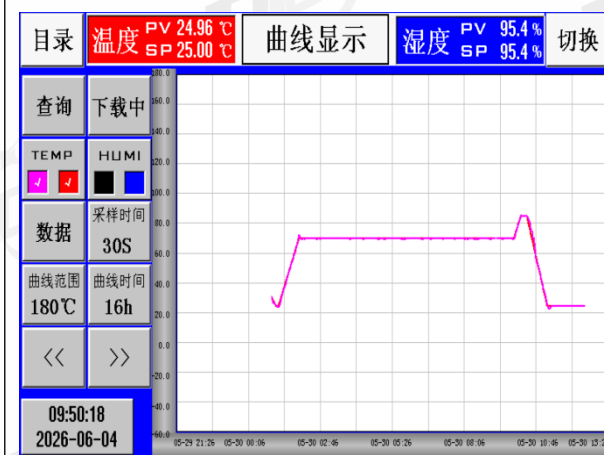




Test setup (2)



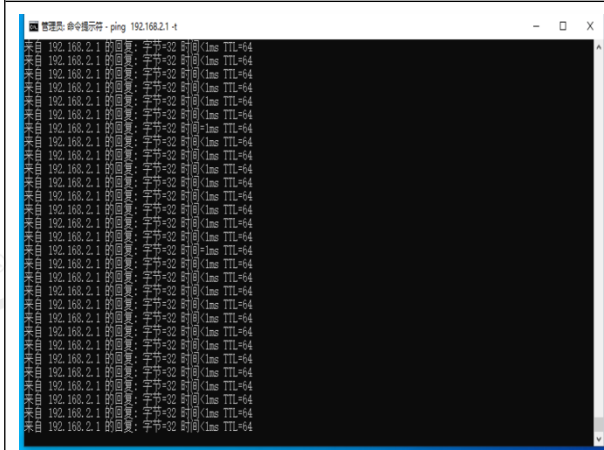
Test curve (10.35V)



Test curve (55.2V)



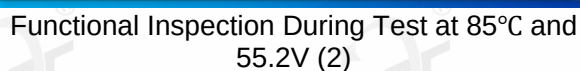
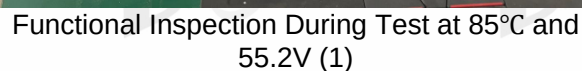
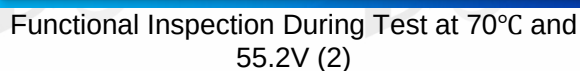
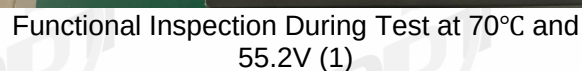
Functional Inspection During Test at 70°C and 10.35V (1)



Functional Inspection During Test at 70°C and 10.35V (2)

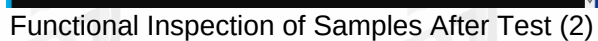
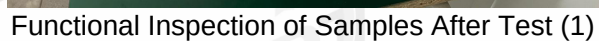


Functional Inspection During Test at 85°C and 10.35V (1)



1





## 4. Low temperature storage Test

### 4.1. Test Equipment

| S/N: | Equipment Name                                                  | Model       | Internal Number | Calibration Validity Period |
|------|-----------------------------------------------------------------|-------------|-----------------|-----------------------------|
| 1    | TEMP&HUMI Programmable chamber                                  | ZXGDJS-225L | DDT-ZC00712     | 2026/09/09                  |
| 2    | High precision DC voltage stable current switching power supply | LW-3020KD   | DDT-ZC04493     | 2026/12/07                  |

### 4.2. Test Conditions

Low temperature: -40°C;  
Duration: 16 h.

### 4.3. Test Requirement

No communication abnormalities shall occur after the test.

### 4.4. Test Sample information

| Sample #. | Sample No.    | Sample Name      | Model No. |
|-----------|---------------|------------------|-----------|
| 3#        | S26041734-003 | InVehicle Router | VR624     |

### 4.5. Test Result

|                                                                 |                                  |
|-----------------------------------------------------------------|----------------------------------|
| Test site: Reliability Laboratory of Songshan Lake Headquarters | Site Ambient: 26.1°C; 58.9%RH    |
| Test Engineer: Yin Xusheng                                      | Test date: 2026/05/30~2026/05/31 |
| Test standard: EN 50155: 2021                                   | Sample #: 3#                     |
| Operating modes: Non-Operating                                  | Sample state during testing: /   |
| Remark: /                                                       |                                  |

No functional abnormalities found after the test.

**Final test conclusion: PASS**

## 4.6. Test Photos

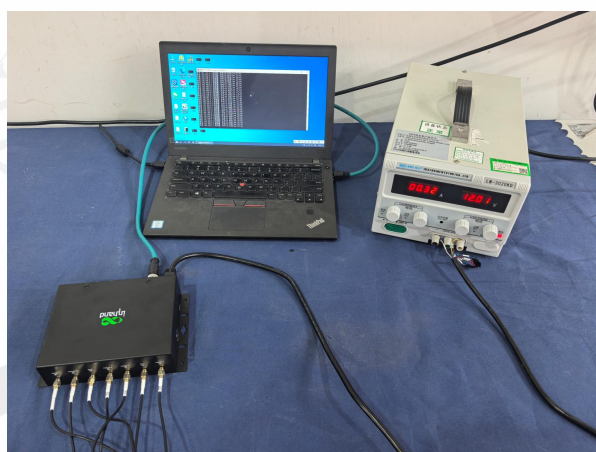
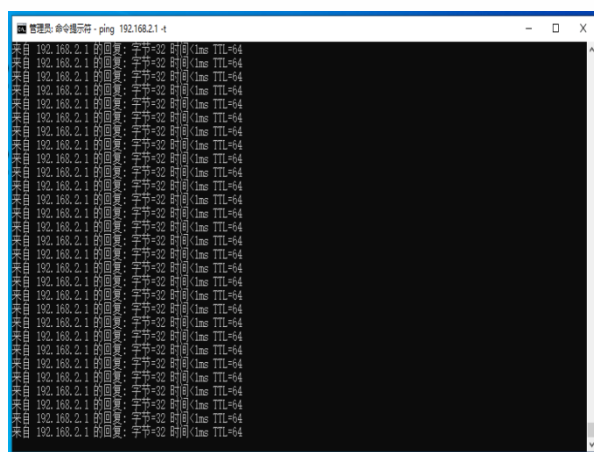
### 4.6.1. Test Photos



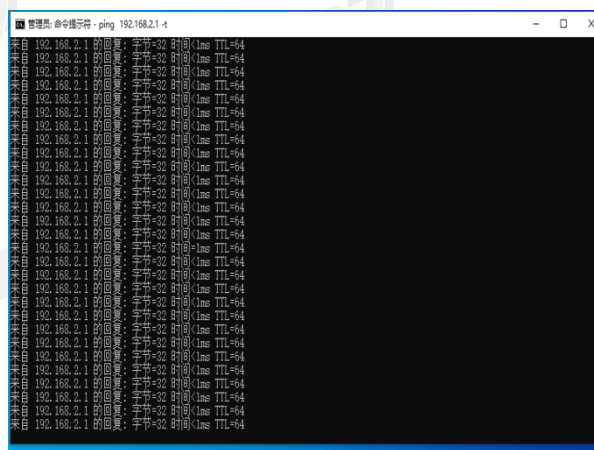
Sample appearance check (1)



Sample appearance check (2)

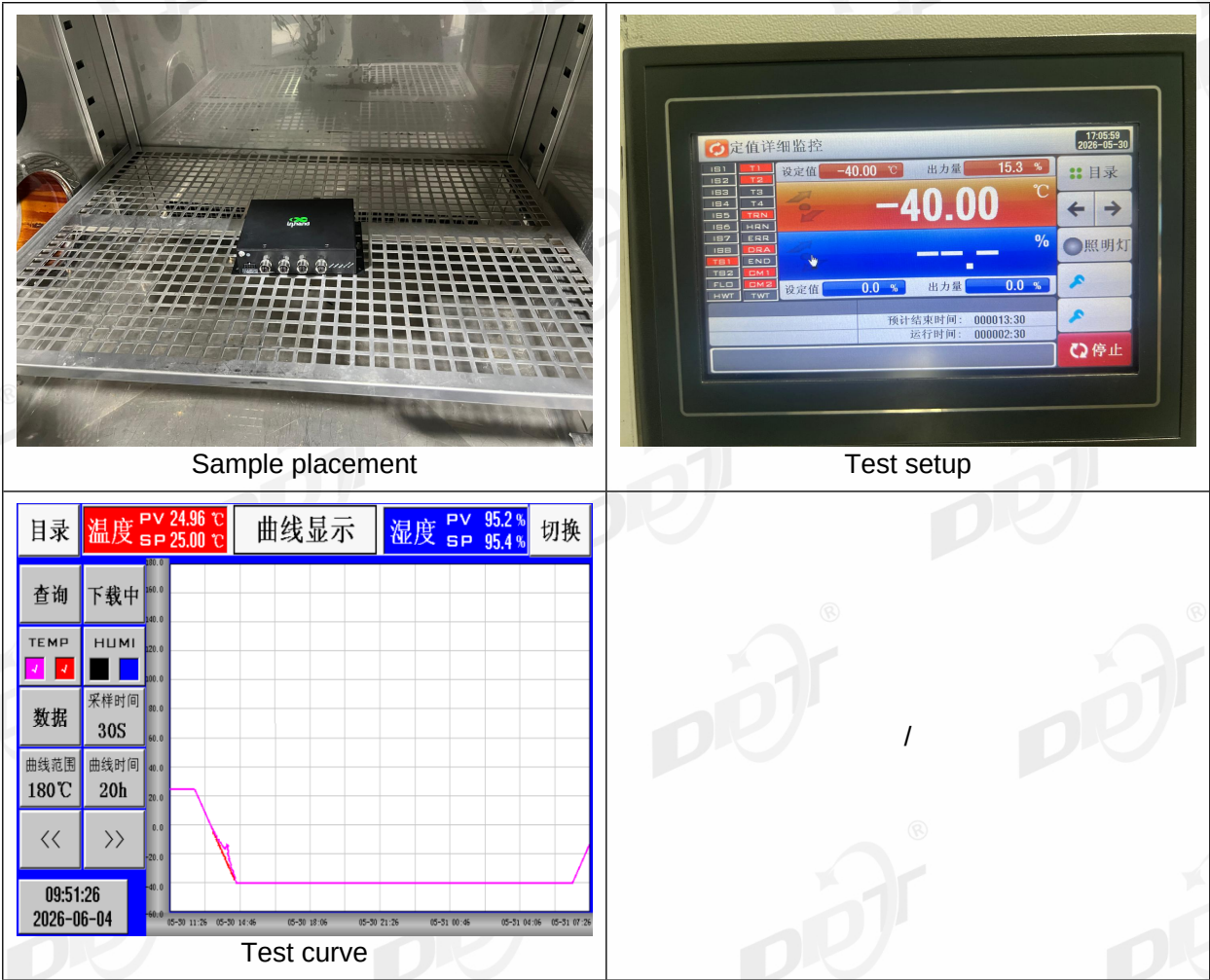
Functional Inspection of Samples Before Test  
(1)Functional Inspection of Samples Before Test  
(2)

Functional Inspection of Samples After Test (1)



Functional Inspection of Samples After Test (2)







## 5. Cyclic damp heat Test

### 5.1. Test Equipment

| S/N: | Equipment Name                                                  | Model       | Internal Number | Calibration Validity Period |
|------|-----------------------------------------------------------------|-------------|-----------------|-----------------------------|
| 1    | TEMP&HUMI Programmable chamber                                  | ZXGDJS-225L | DDT-ZC00712     | 2026/09/09                  |
| 2    | High precision DC voltage stable current switching power supply | LW-3020KD   | DDT-ZC04493     | 2026/12/07                  |

### 5.2. Test Description

|                                                        |                                                                                                                                                              |
|--------------------------------------------------------|--------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Test standard:                                         | EN 50155: 2021                                                                                                                                               |
| Test sample quantity:                                  | 1 PCS                                                                                                                                                        |
| Sample pretreatment description before testing:        | No preprocessing required                                                                                                                                    |
| Sample inspection/testing requirements before testing: | Prior to testing, the specimen shall be inspected for appearance and functions with no abnormalities found.                                                  |
| Sample state during testing:                           | Non-Operating                                                                                                                                                |
| Sample inspection/testing requirements during testing: | Functional check shall be performed for 5 minutes at the start of the second cycle, and the EUT shall remain de-energized for the rest of the test duration. |
| Sample inspection/testing requirements after testing:  | After completion of the test, visual and functional inspection of the equipment under test shall reveal no abnormalities.                                    |

### 5.3. Test Conditions

This test shall be carried out in accordance with EN 60068-2-30:2005 (test Db variant 2); see Figure 17 and Figure 18.

During the 2 cycles of the test, the equipment under test shall not be powered except during the operational check at the beginning of the 2nd cycle.

Temperatures: + 55 °C and + 25 °C

Number of cycles: 2 (breathing effect)

Time: 2 × 24 h

Initial measurement:

- a performance test shall be carried out.

Intermediate measurements:

- at the rise of temperature during the beginning of the 2nd cycle (see Figure 17), during the appearance of condensation on the equipment at 35 °C ± 2 °C, the equipment is temporarily switched ON under nominal voltage and continuous operational checks shall be carried out for a minimum time of 5 min.

At the beginning of the second cycle, if condensation has not occurred (low thermal inertia of equipment) or is not visible, the speed of temperature variation should be increased, but not exceed 1 K/min, at a constant relative humidity level.

Return to room temperature is carried out under controlled recovery conditions.

Check and final measurements:

- continuous operational checks shall be performed at the end of the recovery period, the equipment is switched on and maintained under nominal voltage for the subsequent measurements;
- insulation test shall be performed when equipment is dry (absence of liquid water at the surface of the equipment and associated PBAs), which may take more than 30 min;

— performance test.

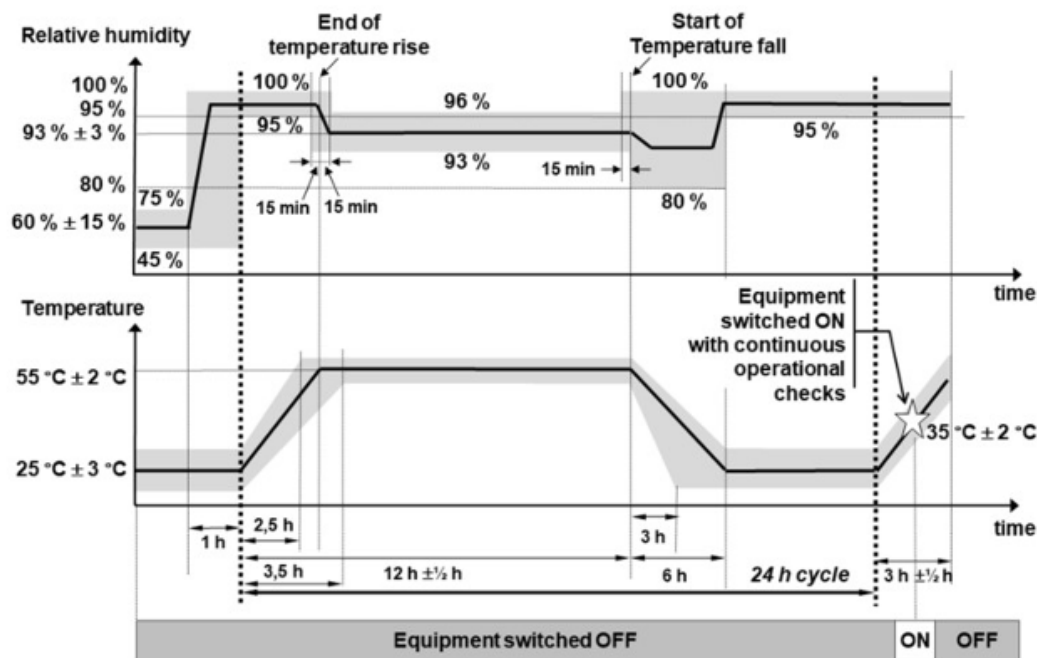


Figure 17 — Cyclic damp heat test: 1<sup>st</sup> and begin of 2<sup>nd</sup> 24-h cycle

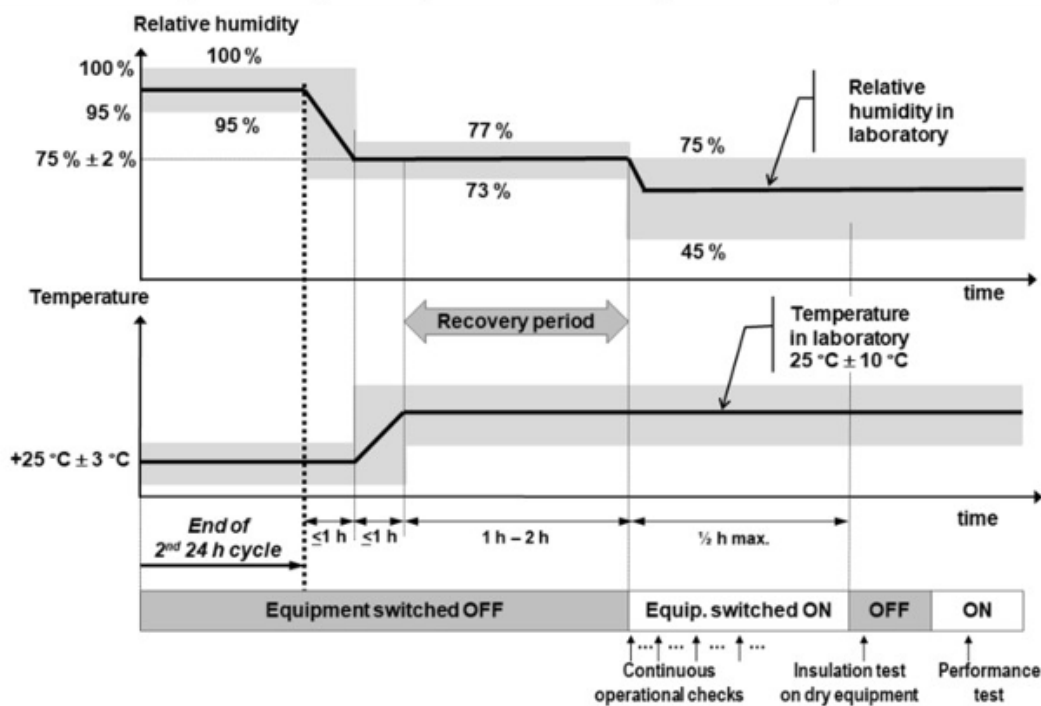


Figure 18 — Cyclic damp heat test: End of 2<sup>nd</sup> cycle and recovery period

NOTE Figure 17 and Figure 18 are taken from EN 60068-2-30:2005, Test Db, Variant 2, added by "Equipment switched ON/OFF" and the different types of tests to be established.

#### 5.4. Test Sample information

| Sample #. | Sample No.    | Sample Name      | Model No. |
|-----------|---------------|------------------|-----------|
| 3#        | S26041734-003 | InVehicle Router | VR624     |

## 5.5.Test Result

| Sample inspection/testing results before testing:          |           |                                               |      |
|------------------------------------------------------------|-----------|-----------------------------------------------|------|
| Inspection/Test date: 2026/06/02                           |           | Test Engineer: Yin Xusheng                    |      |
| Site: Reliability Laboratory of Songshan Lake Headquarters |           | Site Ambient: 24.5°C; 57.7%RH                 |      |
| Item                                                       | Sample #. | Inspection/testing result                     | Memo |
| 1                                                          | 3#        | 1. Appearance normal, 2.Basic function normal | /    |

| Results during testing                                          |                    |           |                                     |      |
|-----------------------------------------------------------------|--------------------|-----------|-------------------------------------|------|
| Test date: 2026/06/02~2026/06/04                                |                    |           | Test Engineer: Yin Xusheng          |      |
| Test site: Reliability Laboratory of Songshan Lake Headquarters |                    |           | Site Ambient: 23°C-28°C,45%RH-75%RH |      |
| Item                                                            | Testing cycle/time | Sample #. | Inspection/testing result           | Memo |
| /                                                               | /                  | /         | /                                   | /    |

|                                                                                   |           |                                               |      |
|-----------------------------------------------------------------------------------|-----------|-----------------------------------------------|------|
| Sample Inspection/Testing results after testing:                                  |           |                                               |      |
| Inspection/Test date: 2026/06/04                                                  |           | Test Engineer: Yin Xusheng                    |      |
| Site: Reliability Laboratory of Songshan Lake Headquarters                        |           | Site Ambient: 25.3°C; 54.3%RH                 |      |
| Sample recovery operation description: Recover to ambient temperature for 2 hours |           |                                               |      |
| Recovery time(H): 2h                                                              |           |                                               |      |
|                                                                                   |           |                                               |      |
| Item                                                                              | Sample #. | Inspection/testing result                     | Memo |
| 1                                                                                 | 3#        | 1. Appearance normal, 2.Basic function normal | /    |

|                                    |
|------------------------------------|
| <b>Final test conclusion: PASS</b> |
|------------------------------------|



## 5.6. Test Photos

### 5.6.1. Before the Test Photos



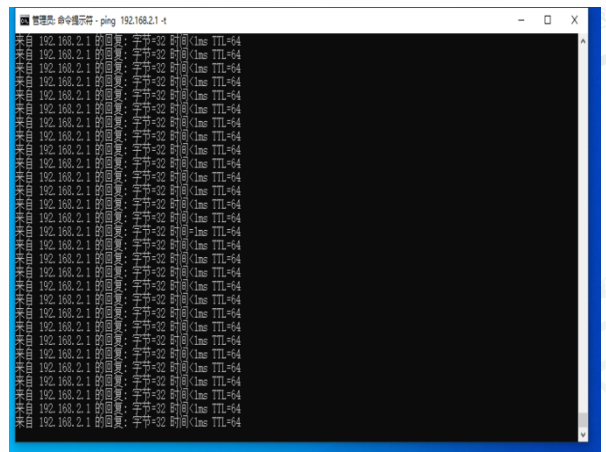
Sample appearance check (1)



Sample appearance check (2)



Sample function check (1)



Sample function check (2)

### 5.6.2. Test Photos



Sample placement

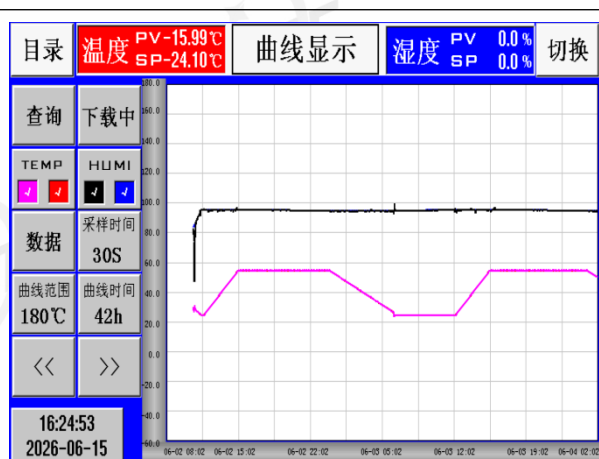


Test setup (1)

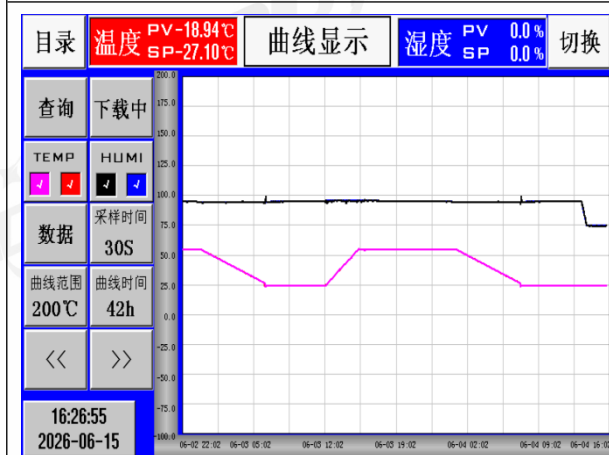




Test setup (2)



Test curve (1)



Test curve (2)

### 5.6.3. After the Test Photos



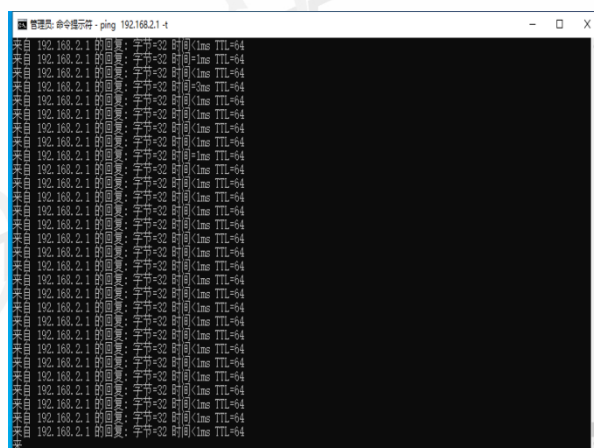
Sample appearance check (1)



Sample appearance check (2)



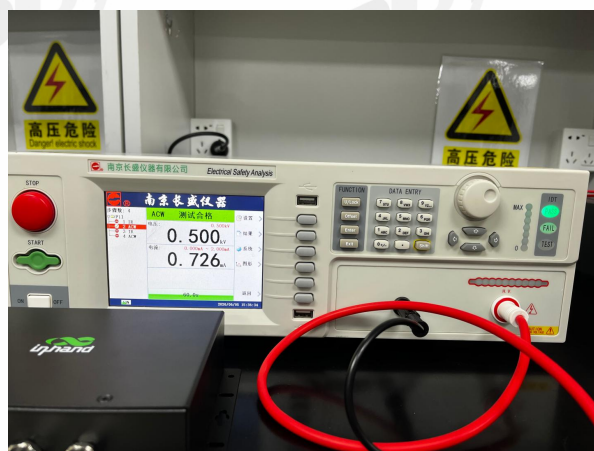
Sample function check (1)



Sample function check (2)



Insulation test (1)



Insulation test (2)

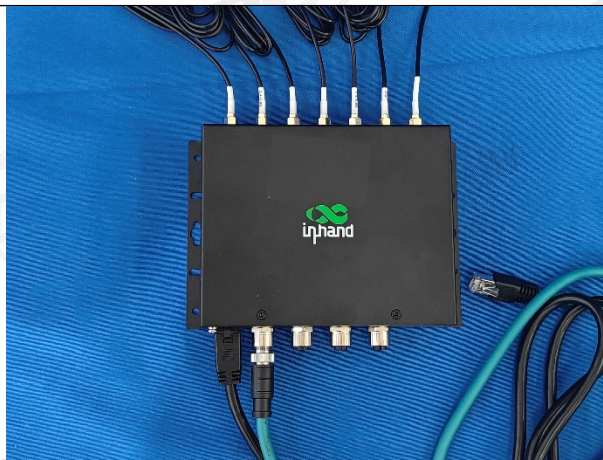


## 6. Visual inspection

### Test conditions:

|                       |                                                                                                                                                                                                                                         |
|-----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Test Standard:        | EN 50155 Section 13.4.1                                                                                                                                                                                                                 |
| Test Procedure:       | The visual inspection verifies the mechanical, dimensional and appearance conformance of the electronic equipment;                                                                                                                      |
| Test sample quantity: | 1PC                                                                                                                                                                                                                                     |
| Test judgment:        | The visual inspection shall be carried out before and after tests to check whether any damage or deterioration has occurred resulting from the tests.                                                                                   |
| Test result:          | Before the test, the appearance of the sample was normal and the function was normal. After the test, the appearance of the mesh disk was normal and it worked normally.<br>The test results complied with the requirement of standard. |

### Test Photo:



Visual Inspection View



Visual Inspection View



Visual Inspection View



Visual Inspection View



Visual Inspection View



Visual Inspection View



## 7. DC Power supply test: supply voltage variations

### Test conditions:

|                       |                                                                                                                                      |
|-----------------------|--------------------------------------------------------------------------------------------------------------------------------------|
| Test Standard:        | EN 50155 Section 13.4.3.2                                                                                                            |
| Test Procedure:       | Tests shall be performed to prove correct functioning at nominal supply voltage and at the specified upper and lower voltage limits. |
| Test sample quantity: | 1PC                                                                                                                                  |
| Test judgment:        | Tests shall be performed to prove correct functioning for the continuous supply voltage range between 0,7 Un and 1,25 Un.            |
| Test result:          | After the test, the prototype functions normally.<br>The test results complied with the requirement of standard.                     |

### Test Equipment:

| Instrument Type        | Internal Number | Model    | Last calibration Date | Next calibration Date |
|------------------------|-----------------|----------|-----------------------|-----------------------|
| Electronic thermometer | DDT-ZC02219     | YSWS3020 | 2025/09/04            | 2026/09/03            |
| DC POWER               | DDT-ZC04962     | PBZ40-10 | 2026/03/19            | 2027/03/18            |

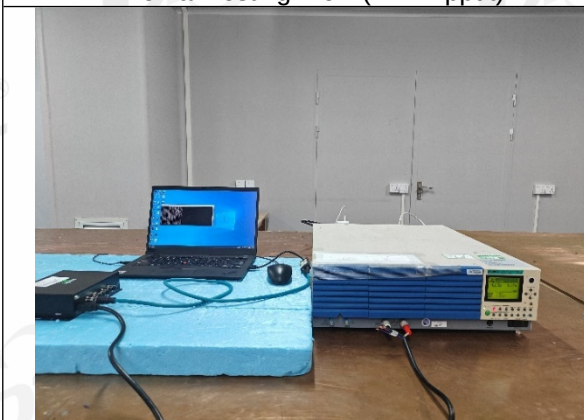
### Test Photo:



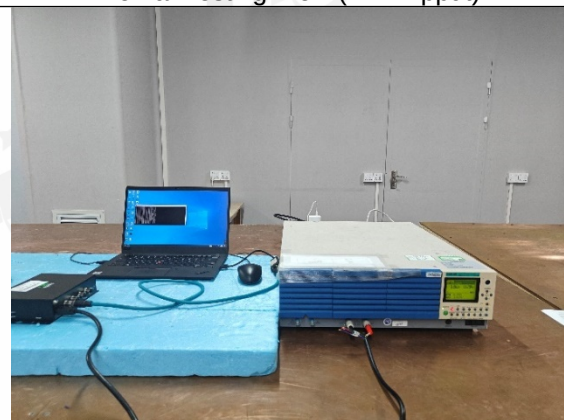
Nomal Testing View (12V inpput)



Nomal Testing View (24V inpput)



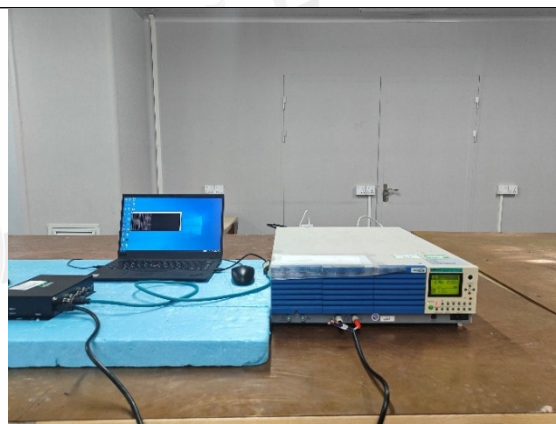
0.7Un Testing View (12V inpput)



0.7Un Testing View (24V inpput)



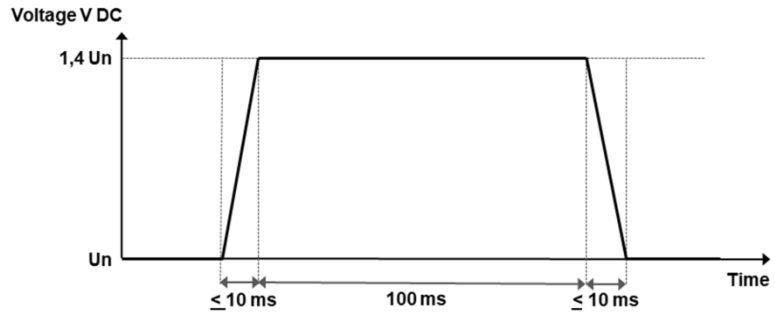
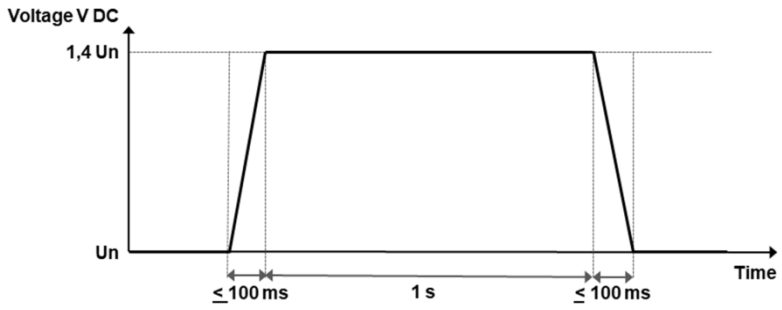
1.25Un Testing View (12V input)



1.25Un Testing View (24V input)

## 8. DC Power supply test: temporary supply overvoltages

### Test conditions:

|                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Test Standard:        | EN 50155 Section 13.4.3.3                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                        |
| Test Procedure:       | <p>Temporary supply overvoltages shall be assumed to be generated with respect to the system power supply return potential.</p> <p>Electronic equipment shall be tested, for each duration (0,1 s and 1 s), with a sequence of 10 overvoltage events. The repetition time should be between 10 s minimum and 1 min maximum (between each test event). If the equipment needs more than 1 min recovery time between test events, this shall be justified in the test specification, noted in the test report and defined in the datasheet of the equipment.</p> <p>Temporary supply overvoltages up to <math>1,4 U_n</math> and not exceeding 0,1 s shall not cause deviation of function (<b>performance criterion A</b>), see Figure 7.</p>  <p><b>Figure 7 — Temporary supply overvoltages up to 0,1 s</b></p> <p>Temporary supply overvoltages up to <math>1,4 U_n</math> and not exceeding 1 s the equipment shall fulfil <b>performance criterion B</b>, see Figure 8.</p>  <p><b>Figure 8 — Temporary supply overvoltages up to 1 s</b></p> <p>If the equipment is specified for a wide input voltage range, temporary supply overvoltage shall be tested according to the highest specified nominal input voltage.</p> <p>EXAMPLE 1 Equipment specified for a grouping of the nominal input voltages 24 V and 28 V: temporary supply overvoltage test is carried out at <math>28 \text{ V} \times 1,4 = 39,2 \text{ V}</math>.</p> <p>EXAMPLE 2 Equipment specified for a grouping of the nominal input voltages from 48 V to 110 V: temporary supply overvoltage test is carried out at <math>110 \text{ V} \times 1,4 = 154 \text{ V}</math>.</p> |
| Test sample quantity: | 1PC                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                              |
| Test judgment:        | Temporary supply overvoltages up to $1,4 U_n$ and not exceeding 0,1 s shall not cause deviation of function (performance criterion A);<br>Temporary supply overvoltages up to $1,4 U_n$ and not exceeding 1 s the equipment shall fulfil performance criterion B, see Figure 8.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Test result:          | After the test, the prototype functions normally.<br>Figure 7 no cause deviation of function (performance criterion A);<br>Figure 8 no cause deviation of function (performance criterion B).<br>The test results complied with the requirement of standard.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

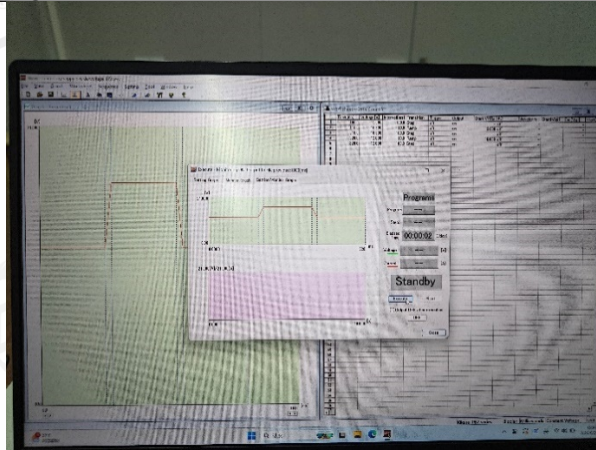


**Test Equipment:**

| Instrument Type        | Internal Number | Model    | Last calibration Date | Next calibration Date |
|------------------------|-----------------|----------|-----------------------|-----------------------|
| Electronic thermometer | DDT-ZC02219     | YSWS3020 | 2025/09/04            | 2026/09/03            |
| DC POWER               | DDT-ZC04962     | PBZ40-10 | 2026/03/19            | 2027/03/18            |

**Test Photo:**

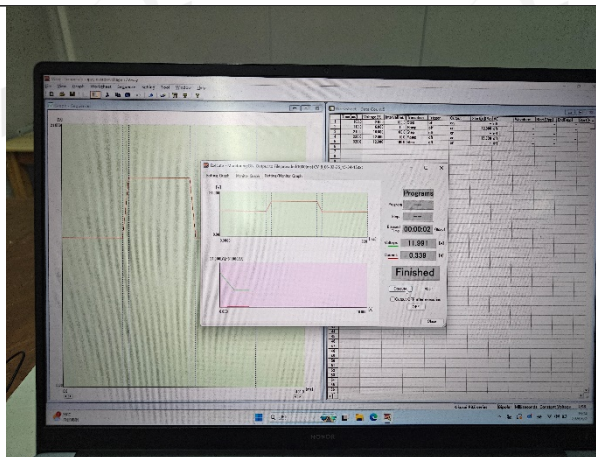
Figure 7 test view



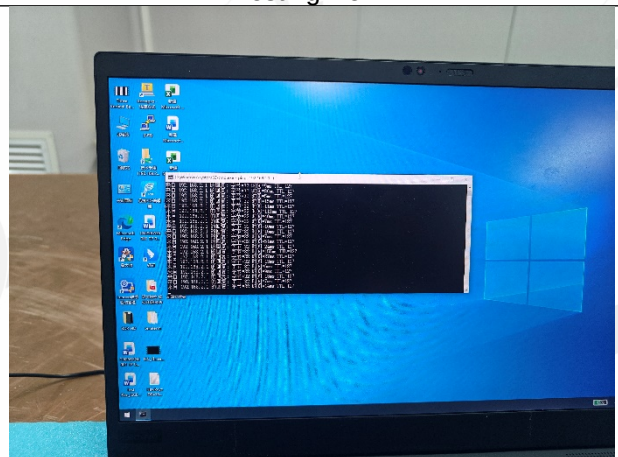
Before test set up view



Testing view

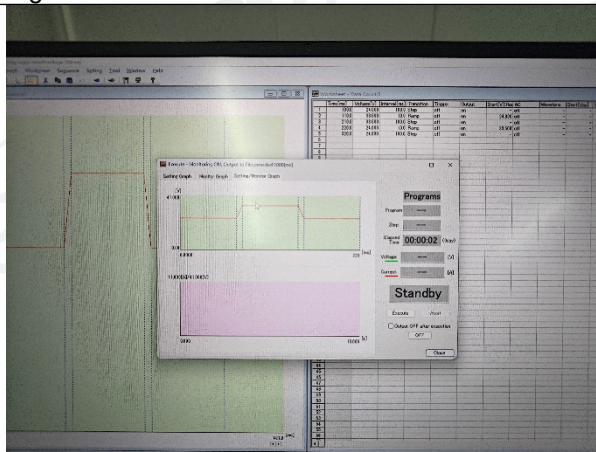


After test view



Function check view

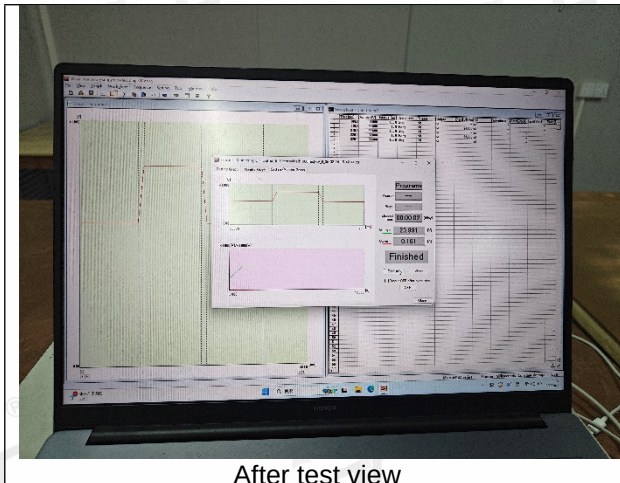
Figure 8 test view



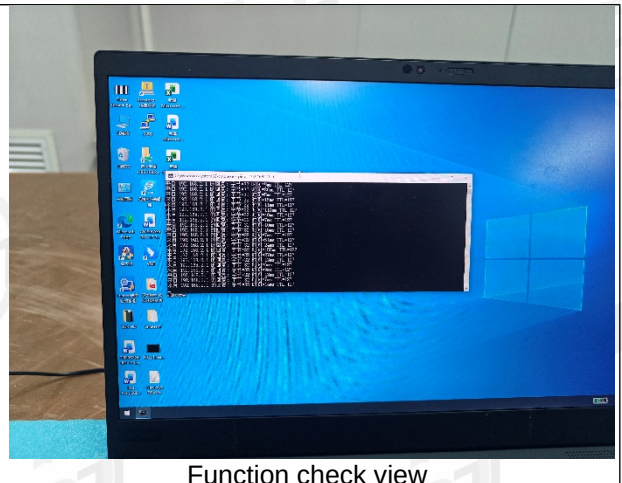
Before test set up view



Testing view



After test view

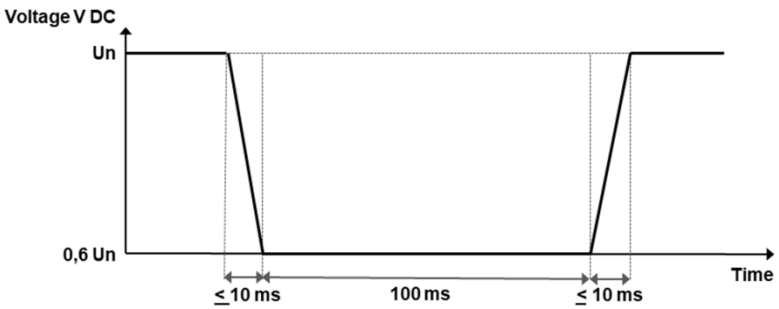


Function check view



## 9. DC Power supply test: temporary supply undervoltage

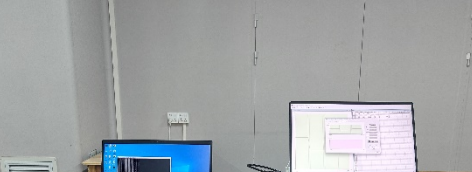
### Test conditions:

|                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                          |
|-----------------------|----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Test Standard:        | EN 50155 Section 13.4.3.4                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| Test Procedure:       | <p>Those tests cover requirements specified in 5.2.3 related to temporary supply undervoltage. Electronic equipment shall be tested, for the duration of 0,1 s, with a sequence of 10 undervoltage events. The repetition time should be between 10 s minimum and 1 min maximum (between each test event). If the equipment needs more than 1 min recovery time between test events, this shall be justified in the test specification, noted in the test report and defined in the datasheet of the equipment.</p> <p>Temporary supply undervoltage down to <math>0,6 U_n</math> and not exceeding 0,1 s shall not cause deviation of function (<b>performance criterion A</b>), see Figure 9.</p>  <p><b>Figure 9 — Temporary supply undervoltage</b></p> <p>If the equipment is specified for a wide input voltage range, temporary supply undervoltage shall be tested according to the lowest specified nominal input voltage.</p> <p>EXAMPLE 1 Equipment specified for a grouping of the nominal input voltages 24 V and 28 V: temporary supply undervoltage test is carried out at <math>24 \text{ V} \times 0,6 = 14,4 \text{ V}</math>.</p> <p>EXAMPLE 2 Equipment specified for a grouping of the nominal input voltages from 48 V to 110 V: temporary supply undervoltage test is carried out at <math>48 \text{ V} \times 0,6 = 28,8 \text{ V}</math>.</p> |
| Test sample quantity: | 1PC                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| Test judgment:        | Temporary supply undervoltage down to $0,6 U_n$ and not exceeding 0,1 s shall not cause deviation of function (performance criterion A)                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                  |
| Test result:          | After the test, the prototype functions normally.<br>Figure 9 no cause deviation of function (performance criterion A);<br>The test results complied with the requirement of standard.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |

### Test Equipment:

| Instrument Type        | Internal Number | Model    | Last calibration Date | Next calibration Date |
|------------------------|-----------------|----------|-----------------------|-----------------------|
| Electronic thermometer | DDT-ZC02219     | YSWS3020 | 2025/09/04            | 2026/09/03            |
| DC POWER               | DDT-ZC04962     | PBZ40-10 | 2026/03/19            | 2027/03/18            |

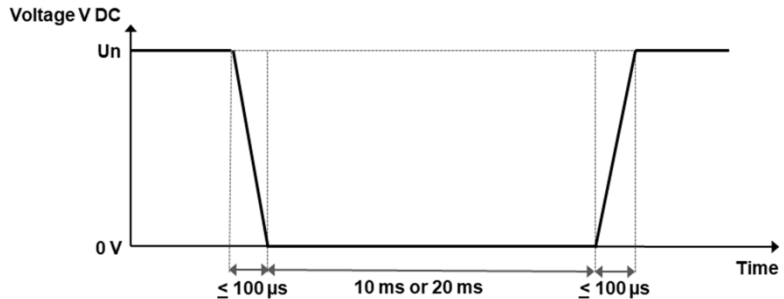




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## 10. DC Power supply test: interruptions of supply voltage

### Test conditions:

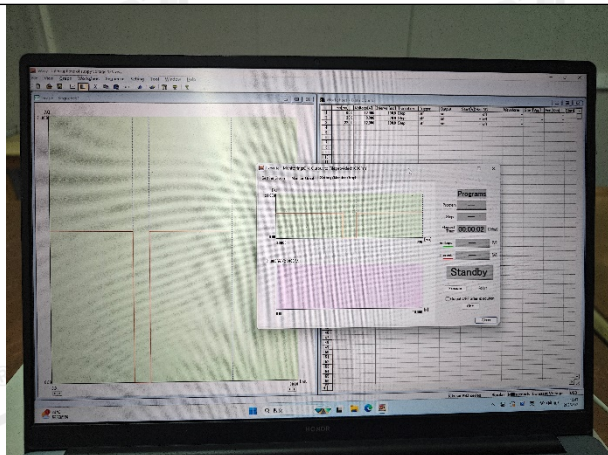
|                       |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Test Standard:        | EN 50155 Section 13.4.3.5                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                           |
| Test Procedure:       | <p>To verify that performance criterion A is fulfilled, the maximum interruption time for the relevant class shall be applied. Electronic equipment shall be tested with a sequence of 10 interruption events. The repetition time should be between 10 s minimum and 1 min maximum (between each test event).<br/>If the equipment needs more than 1 min recovery time between test events, this shall be justified in the test specification, noted in the test report and defined in the datasheet of the equipment.<br/>To verify that performance criterion C is fulfilled it is suggested to test with different interruption times greater than the time given in the relevant class.<br/>Tests shall be carried out at nominal voltage and nominal load / operating conditions.</p>  <p>Figure 10 — Interruption of supply voltage</p> <p>If the equipment is specified for a wide input voltage range, interruption of supply voltage shall be tested at the lowest specified nominal input voltage.</p> <p>EXAMPLE 1 Equipment specified for a grouping of the nominal input voltages 24 V and 28 V: interruption test of supply voltage is carried out at 24 V.</p> <p>EXAMPLE 2 Equipment specified for a grouping of the nominal input voltages from 48 V to 110 V: interruption test of supply voltage is carried out at 48 V.</p> |
| Test sample quantity: | 1PC                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Test judgment:        | Electronic equipment shall be tested with a sequence of 10 interruption events. The repetition time should be between 10 s minimum and 1 min maximum (between each test event). Verify that performance criterion A                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| Test result:          | After the test, the prototype functions normally.<br>The test results complied with the requirement of standard.                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |

### Test Equipment:

| Instrument Type        | Internal Number | Model    | Last calibration Date | Next calibration Date |
|------------------------|-----------------|----------|-----------------------|-----------------------|
| Electronic thermometer | DDT-ZC02219     | YSWS3020 | 2025/09/04            | 2026/09/03            |
| DC POWER               | DDT-ZC04962     | PBZ40-10 | 2026/03/19            | 2027/03/18            |



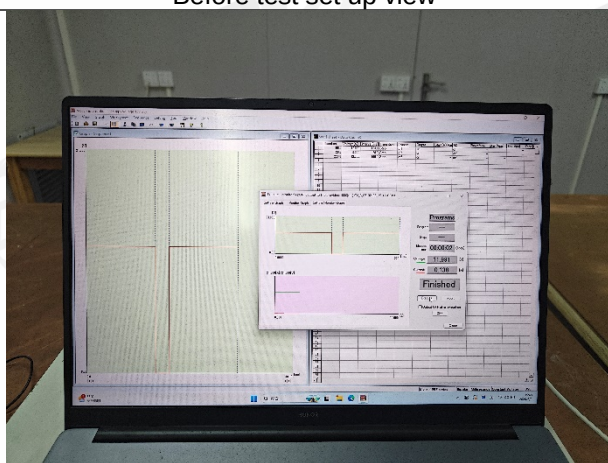
## Test Photo:



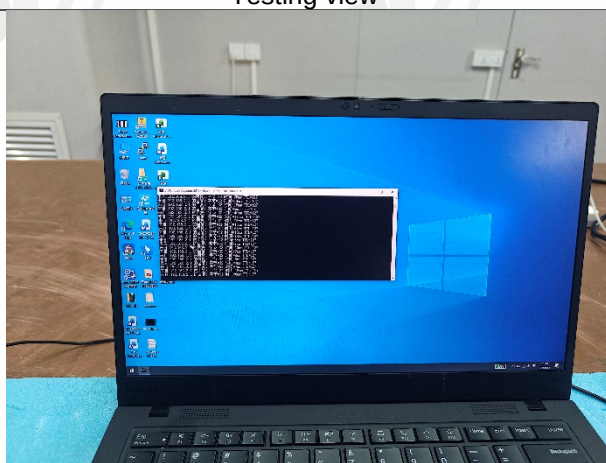
Before test set up view



Testing view



After test view



Function check view



## 11. Insulation test

### Test Equipment:

| Instrument Type            | Internal Number | Model    | Last calibration Date | Next calibration Date |
|----------------------------|-----------------|----------|-----------------------|-----------------------|
| Electronic thermometer     | DDT-ZC02219     | YSWS3020 | 2025/09/04            | 2026/09/03            |
| Dielectric Strength Tester | DDT-ZC01459     | TOS9201  | 2025/12/30            | 2026/12/29            |

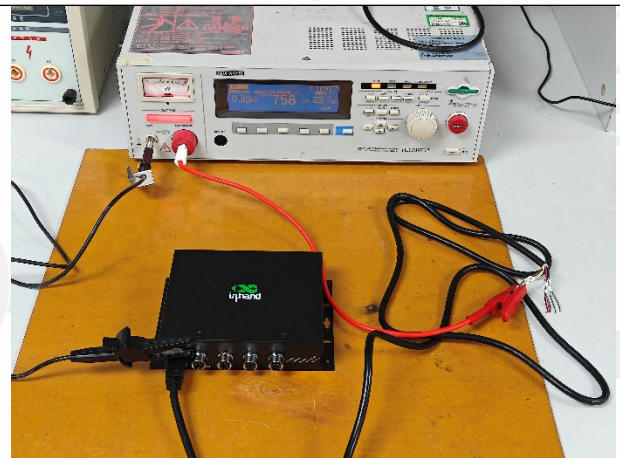
### Test conditions:

|                       |                                                                                                                                                                                     |
|-----------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Test Standard:        | EN 50155 Section 13.4.7                                                                                                                                                             |
| Test Procedure:       | 13.4.7.3 Voltage withstand test:<br>Test Voltage: 500Vdc;<br>Test time: 60S;<br>Rise time: 10S;<br>13.4.7.2 Insulation resistance test:<br>Test Voltage: 500Vdc;<br>Test time: 60S; |
| Test sample quantity: | 1PC                                                                                                                                                                                 |
| Test judgment:        | Electric strength test: No breakdown;<br>Insulation test: $\geq 20M\Omega$ .                                                                                                        |
| Test result:          | The test results complied with the requirement of standard.                                                                                                                         |

### Test Photo:



Before Voltage withstand test set up view



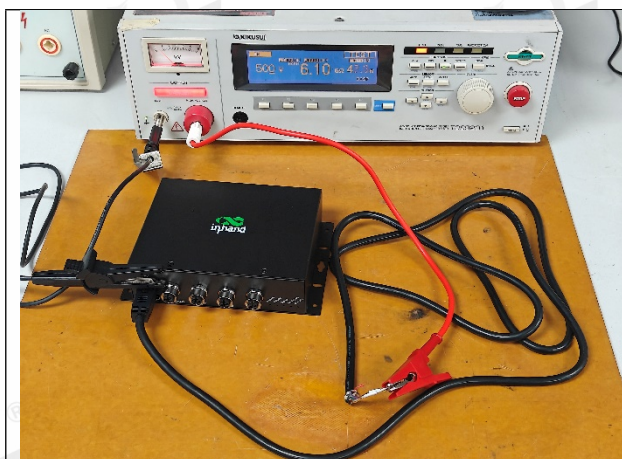
Voltage withstand Testing view



After Voltage withstand test view



Before Insulation resistance test set up view



Insulation resistance Testing view



After Insulation resistance test view

-----End Report-----